Anja Šurina

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Education

EPFL Switzerland | Sep 2023 – Present
PhD in Computer and Communication Sciences
- Research focus: LLM post-training and test-time scaling for scientific and mathematical discovery
- Advisors: Prof Caglar Gulcehre, Prof Emmanuel Abbe

ETH Zurich & University of Zurich Switzerland | Sep 2020 – May 2023
MSc in Neural Systems and Computation
- Focus: Artificial Intelligence, Deep Learning, Computational Neuroscience – GPA 5.8/6.0
- Thesis on meta continual learning under distribution shifts, advised by Dr João Sacramento

University of Ljubljana Slovenia | Oct 2017 – Jun 2020
BSc in Electrical Engineering
- Focus: Information and Communications Technology – GPA 9.7/10.0, ranked 1st of 165 students

Research Experience

Inherent Laboratories United Kingdom | Jun 2026 – Present
Research Fellow
- Long-horizon RL post-training of an AI agent for automated scientific research
- Advisor: Dr Edward Hughes

Google DeepMind United Kingdom | Aug 2025 – Apr 2026
Student Researcher, AlphaEvolve and AlphaProof Nexus teams
- AI agents for formal proof search in Lean: resolved open Erdős problems and advanced research mathematics
→ Paper in *Science 2026*
- Advisor: Dr Francisco J. R. Ruiz

Mila – Quebec AI Institute Canada | Jun 2023 – Aug 2023
Visiting Student Researcher
- Design factors for training neural networks with GFlowNet objectives
- Advisor: Prof Yoshua Bengio

Stanford University United States | Oct 2022 – Dec 2022
Visiting Student Researcher
- Meta-learning for biomedical zero-shot causal inference: effects of novel drugs and cell perturbations
→ Paper at *NeurIPS 2023 (Spotlight)*
- Advisor: Prof Jure Leskovec

IBM Research Zurich Switzerland | Mar 2021 – Jan 2022
Research Intern, Emerging Computing and Circuits Group
- Bio-inspired online learning for artificial spiking neural networks on neuromorphic hardware
→ Paper at *IEEE AICAS 2022*
- Advisors: Prof Giacomo Indiveri, Dr Angeliki Pantazi

Publications

Training AI Scientists to Replicate Research
Damon Falck, Samer Sabri, **Anja Surina**, Thom Foster, Anya Sims, Sam Devlin, Dylan Rogers, Tantum Collins, Kaloyan Aleksiev, Louis Kirsch, Edward Hughes
arXiv 2026 2608.13331

The Future of Facts: Tracing the Factual Generation-Verification Gap
Tim R. Davidson*, **Anja Surina***, Caglar Gulcehre
NeurIPS 2026 Neural Information Processing Systems
COLM 2026 ***Best Paper Award*** Conference on Language Modeling, Scientific Understanding of Foundation Models Workshop

Advancing Mathematics Research with AI-Driven Formal Proof Search

George Tsoukalas, Anton Kovsharov, Sergey Shirobokov, **Anja Surina**, Moritz Firsching, Gergely Bérczi, Francisco J. R. Ruiz, Arun Suggala, Adam Zsolt Wagner, Eric Wieser, Lei Yu, Aja Huang, Miklós Z. Horváth, Andrew Ferraiuolo, Henryk Michalewski, Codrut Grosu, Edward Lockhart, Thomas Hubert, Matej Balog, Pushmeet Kohli, Swarat Chaudhuri

Science 2026

An Improved Last-Iterate Convergence Rate for Anchored Gradient Descent Ascent

Anja Surina, Arun Suggala, George Tsoukalas, Anton Kovsharov, Sergey Shirobokov, Francisco J. R. Ruiz, Pushmeet Kohli, Swarat Chaudhuri

arXiv 2026 2604.03782 Results later incorporated into the **Science** paper above

Algorithm Discovery With LLMs: Evolutionary Search Meets Reinforcement Learning

Anja Surina, Amin Mansouri, Lars Quaedvlieg, Amal Seddas, Maryna Viazovska, Emmanuel Abbe, Caglar Gulcehre

COLM 2025 Conference on Language Modeling

ICLR 2025 International Conference on Learning Representations, Scaling Self-Improving Foundation Models Workshop

The Role of Deep Learning Regularizations on Actors in Offline RL

Denis Tarasov, **Anja Surina**, Caglar Gulcehre

arXiv 2024 2409.07606

Zero-Shot Causal Learning

Hamed Nilforoshan, Michael Moor, Yusuf Roohani, Yining Chen, **Anja Surina**, Michihiro Yasunaga, Sara Oblak, Jure Leskovec

NeurIPS 2023 ***Spotlight*** Neural Information Processing Systems

Biologically-Inspired Training of Spiking Recurrent Neural Networks with Neuromorphic Hardware

Thomas Bohnstingl, **Anja Surina**, Maxime Fabre, Yiğit Demirağ, Charlotte Frenkel, Melika Payvand, Giacomo Indiveri, Angeliki Pantazi

AICAS 2022 IEEE International Conference on Artificial Intelligence Circuits and Systems

* equal contribution

Awards & Honours

Forbes 30 Under 30 Slovenia, Science & Technology category — 2025
EPFL EDIC Doctoral Fellowship, first-year fellowship for research rotations — 2023 – 2024
ASEF Fellowship, funded research visit at Stanford University — 2023
Ad Futura Scholarship, for graduate studies abroad — 2020 – 2022
Rector's Honour List of Exceptional Students of the University — 2020
Dean's Award for Excellence upon Graduation — 2020
Zois Scholarship, national merit scholarship — 2017 – 2020

Teaching

Teaching Assistant, **Reinforcement Learning**, EPFL — 2024, 2025

Other Activities

Leadership: Elected President of the Student Council at the Faculty of Electrical Engineering — 2019 – 2020
Sports: Bronze medal at the IDO World Dance Championship, multiple national titles — 2012 – 2018